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Contour Detection based Ear Recognition for Biometric Applications

Mangayarkarasi N[a,*], Raghuraman G[a], Nasreen A[a]

[a]*SSN College of Engineering,Chennai,India,603110*

Abstract

Now-a-days, securing ones information has become a greatest threat. The advancement in the technology has paved way for many biometrics systems. These biometrics using finger prints have been highly hacked these days. This paper suggests a technique using the human ear as the perfect source of person identification as it has its own unique features. A few image processing algorithms were applied to segment the ear but some lacked in perfect. The proposed system emphasizes on developing a biometrics system that would detect and recognize a person using the features of their ear where the ear is segmented using appropriate segmentation techniques. The ear recognition is then carried out by using feature matching technique.

Keywords: *Secure data;biometrics;huma ear;contour detection; SIFT features;ear recognition.*

1. Introduction

Ear is one of the most suitable candidate to be used for biometrics. It does not change with emotions, states of mind, sadness, fear or cosmetic changes. Ear can be easily captured at a distance, even if the subject is not fully

* Corresponding author. Tel.:+91-8056421789
E-mail address: mangayarkarasin@ssn.edu.in

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cooperative. This makes ear recognition especially interesting for smart surveillance tasks and for forensic image analysis as well. Ear does not have a completely random structure. It has standard parts as other biometric traits like face. Unlike human face, ear has no expression changes or make-up. Ear also has other prominent features [1] such as the anti-helix which runs parallel to the helix, and a distinctive hairpin-bend shape just above the lobe called the inter tragic notch. The central area or concha is named for its shell-like appearance.

2. Related Work

Ear detection is one of the important steps for recognition. Various image processing algorithms that are used for detection of human ear include: snake algorithm [1], 3D reconstruction from 2D images. In order to recognize the ear, the classification algorithms are applied. For this, initially the feature extraction techniques like: geometrical feature extraction, Gabor transform and force field transform are used. The conventional machine learning algorithms that are applied include: Fisher discriminant analysis [5], neural network [8], Metaface dictionary learning [4] and Sparse Representation based Classification (SRC).

In [1], shape based classification approach is proposed. Initially the images are resized and the ear is identified using the snake algorithm. Then, the geometrical features [3] like: mean, centroid, Euclidean distance and shape are extracted and classification is done using K- nearest neighbor (KNN) algorithm. [2] Suggested a 3D profile based method for localizing the ear. The depth maps [6-7] of the images are created and the edge information in this map is used to construct the graph with connected component analysis. In [3], ear recognition for occluded images is proposed by using the sparse representation based classification. The Gabor features are extracted for occluded and non occluded images.

The existing methods lacked in proposing segmentation based ear recognition system and also were not consistent for different scales and angles of the human ear. So to make the system work irrespective of the size and position or angle of the ear, Scale Invariant Feature Transform (SIFT) method [9] has been used. Thus this paper focuses on recognizing a person using his/her ear. As the features of the ear for a person varies from the other, there are certain features of the ear deciding the uniqueness of an ear that include: concha, helix and anti-helix. These feature values are estimated using the Scale Invariant Feature Transform (SIFT) method and the person is recognized by finding the Euclidean distance between the trained and tested feature vectors. The set with minimum distance shows that they are more similar and thus a person is recognized.

3. Proposed work

The system architecture shown in Fig 1 gives an overall view about all the modules in the proposed system and the flow of the process right from data collection to classification. Initially the human ear images of different people are captured and stored in the database. Also the various image dataset of the human ear have been collected from the online blog [10]. This data set consists of the images of right ear, left ear, front-view, down view and back view of each person's ear. All the images are of size 492x702. Next, the preprocessing step involves conversion of the RGB image to gray scale image. This gray image is given as an input to the contour detection algorithm. To detect the ear, here the masks are initially designed for the ear image by setting a threshold value. This mask helps in segmenting the ear out from its background elements including hair. By convolving the mask and the gray image we get the segmented final mask that is in binary format. This final mask is then applied over the input image and the bounding box is drawn around the ear part. By knowing the dimension of the bounding box, the ear is cropped and saved for the use of feature extraction. This image is then given as the input for feature extraction phase where the SIFT algorithm is applied and finally the SIFT feature vectors are found. These feature vectors obtained from the image of each person in the training phase are stored for the latter use. In the testing phase, when a random person enters his/her ear image is captured. This input test image undergoes all the process as that of training images.

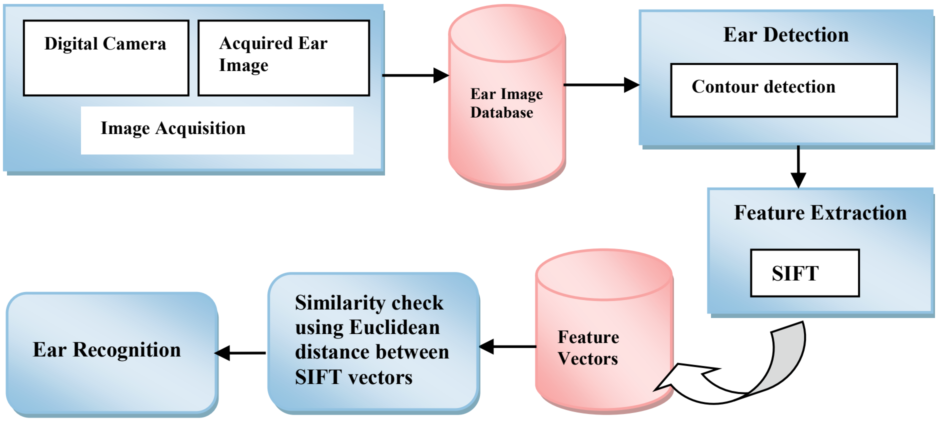

Fig 1. System architecture for ear recognition

Finally the SIFT vectors are got. By using the Euclidean distance measure method, the person is recognized by evaluating the SIFT vectors from the already stored SIFT values which were obtained from training set and the SIFT vectors obtained from testing set. If the distance between the two vectors is minimum then it shows that they are similar and thus, every person is recognized by using this value.

3.1. Image Acquisition

The human ear dataset has been created for this work. The first stage of any vision system is the image acquisition stage. After the image has been obtained, various methods of processing are applied to the image to perform the many different vision tasks required today. However, if the image has not been acquired satisfactorily then the intended tasks may not be achievable, even with the aid of some form of image enhancement. Ear image of about 25 people have been collected irrespective of the age. Dataset has both left and right images of the human ear. Other than this, the online dataset of the human ear is also used.

3.2. Pre-processing

Original image is resized to [492x702] pixels to be same as the size of images in database and the resized images are converted into gray scale images.

3.3. Snake Algorithm

Active contour model, also called snakes, is a framework in computer vision for delineating an object outline from a possibly noisy 2D image. The snake model is popular in computer vision, and snakes are greatly used in applications like object tracking, shape recognition, segmentation, edge detection and stereo matching. A snake is an energy minimizing, deformable spline influenced by constraint and image forces that pull it towards object contours and internal forces that resist deformation. In two dimensions, the active shape model represents a discrete version of this approach, taking advantage of the point distribution model to restrict the shape range to an explicit domain learned from a training set.

3.4. Drawbacks of Snake Algorithm

Snakes do not solve the entire problem of finding contours in human, since the method requires knowledge of the desired contour shape beforehand. Rather, they depend on other mechanisms such as interaction with a user, interaction with some higher level image understanding process, or information from image data adjacent in time or space. Each ear has its own threshold. A common threshold point cannot be fixed for ear in general. So we go for canny edge detector.

3.5. Proposed Contour detection algorithm

There were certain drawbacks inferred from the snake algorithm which includes different threshold for each ear image and the exact contour of the ear wasn't been able to obtain. The contour detection of the human ear is the important phase because only if the ear part of the person is segmented properly from the other parts like skin and hair, the features extraction results will be accurate. The accuracy of the feature vector values also decides the correct recognition of the person's ear. The proposed contour detection algorithm includes the following steps:

1. Initially, a mask is formed which contains only the 0s and 1s.This mask is of the size of the input image.
2. The mask has the 1s values from the threshold of 150 and above, till the maximum size of the image.
3. The next step is to convolute the mask with the input gray image. A binary output of the segmented ear image is displayed.
4. This is then applied to the original RGB image to detect only the ear part. A bounding box is drawn around the ear.
5. Now, by knowing the size of the bounding box, the ear part is alone segmented out and saved.

3.6. Scale Invariant Feature Transform (SIFT) Algorithm

SIFT algorithm is used to detect and describe the image local features like rotation, scale and viewpoints in computer vision. It has key features as the output to detect the ear. It is more effective one for the recognition process. By having this test SIFT vectors, the exact match is found from the already stored SIFT values from training set by using the Euclidean distance measure method. There are certain steps involved in extracting the SIFT features which include:

1. *Constructing a scale space*- Here original image is taken and it generates the progressively blurred out images. Next the original image is resized to half size. Again it generates blurred out images and it keeps repeating.
2. *Laplacian of Gaussian (LoG) Approximation* – It helps in locating the edges and corner of the ear image. The Laplacian pyramid is build initially, and then the Gaussian operator is applied to blur the unwanted noises.
3. *Finding the key points* – This is the most important step because: the key points are regarded as the unique features that are helpful to distinguish a person's ear with that of the other person. The unique features with respect to human ear include the regions like concha, helix and anti-helix. The key points are found by: i) locating the maxima/minima in Difference of Gaussian (DoG) images and ii) finding the sub pixel maxima/minima.
4. *Get rid of bad key points* - Sometimes the low contrast region and edges are identified as the key points. So to remove these bad key points, Harris corner detection method is applied.
5. *Assigning an orientation to the key points* – Each key points has a different magnitude and angle. These two factors are calculated for every key point.
6. *Generate SIFT features* - A 16x16 window is considered around the key points which is further broken into 4x4 windows. The magnitude and orientation are calculated within each window and are put into an 8 bin histogram. Finally on covering the entire 4x4 sub windows: 4x4x8= 128 SIFT feature vectors are generated.

4. Results and discussion

4.1. Results of Snake Algorithm

The input image containing the human ear is given as the input to the snake algorithm. The initial mask is formed from the binarized input image. Then the outer boundary mask is formed and by setting the threshold value (T), the contour of the ear is drawn. For this the different threshold values are set so that when the mask is applied over the input image, it converges towards the boundary of the ear. The output obtained from snake algorithm using the threshold value T=80, 90, 95 and 101 are shown in Fig 2. By convoluting the inner and the outer boundary mask of the ear over the gray scale image, the contour is drawn over the original image. Once the snake finds the contour, the ear is detected.

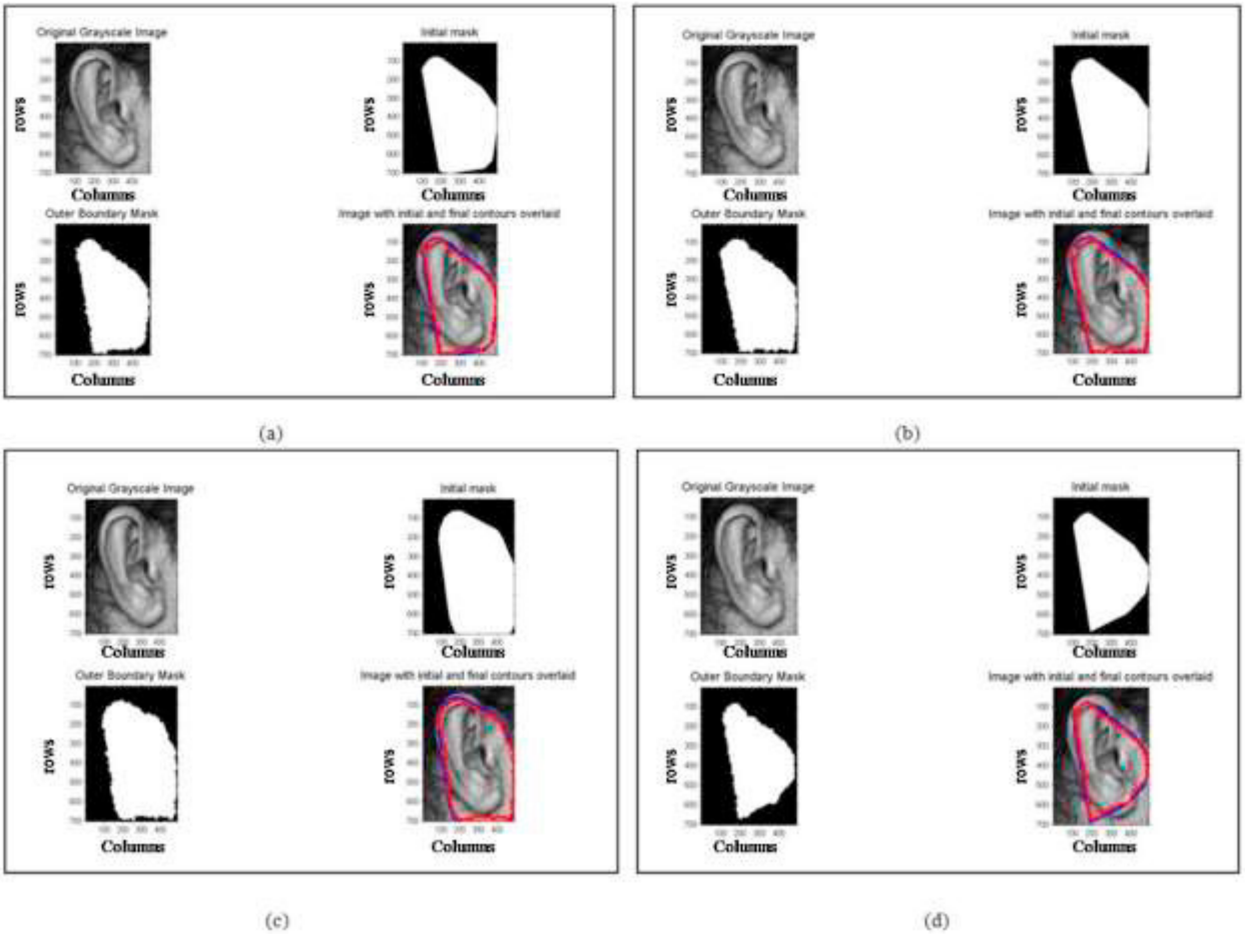

Fig 2. Contour detection using Snake algorithm with thresholds (T) (a) T=80, (b) T=90, (c) T=95, (d) T=101

Therefore, on implementation we inferred that on varying the threshold value of the mask from 80 and 110, the contour detection gave better result. But it has also been inferred that the entire part of the ear has not been covered by the contour. So the snake becomes less flexible for this particular problem of ear detection. So, based on these observations a novel contour detection algorithm has been proposed which is discussed in next part.

4.2. Results of Contour Detection and SIFT Algorithm

Since the above method did not give accurate results, a novel contour detection method is proposed. In this, the preprocessing involves the conversion of the RGB image to gray scale image. Now, a mask is formed which contains only the 0's and 1's as shown in Fig 3. This mask is of the size of the input image. The mask has the 1's values from the threshold of 150 and above, till the maximum size of the image.

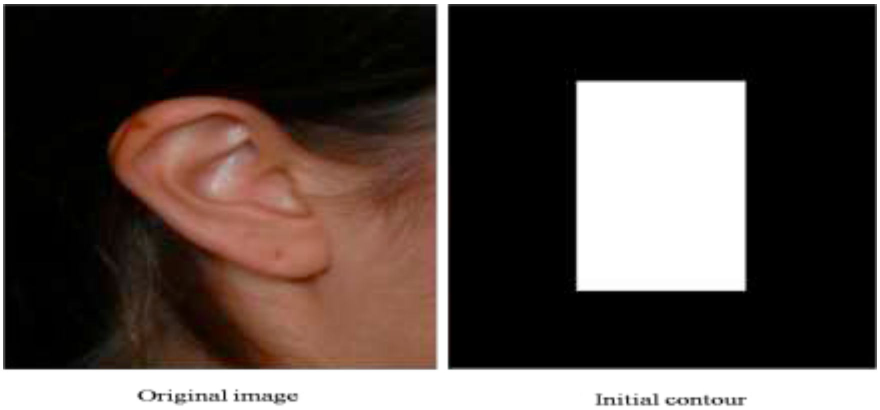

Fig 3. Assigning the initial contour for detection

The next step is to convolute the mask with the input gray image. Thus, we get a binary output of the segmented ear image as in Fig 4. This is then applied to the original RGB image to detect only the ear part.

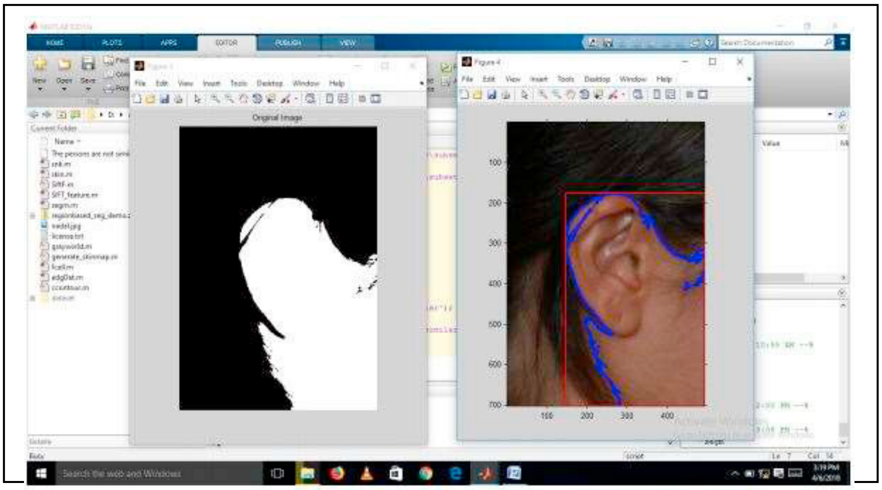

Fig 4. Ear detection using the proposed contour detection algorithm

A bounding box is drawn around the ear. Now, by knowing the size of the bounding box, the ear part is alone segmented out as in Fig 5 and saved for the later use. The segmented image is then given as the input for feature extraction phase. Here the SIFT features are extracted as in Fig 6, by employing the steps as discussed in the previous chapter.

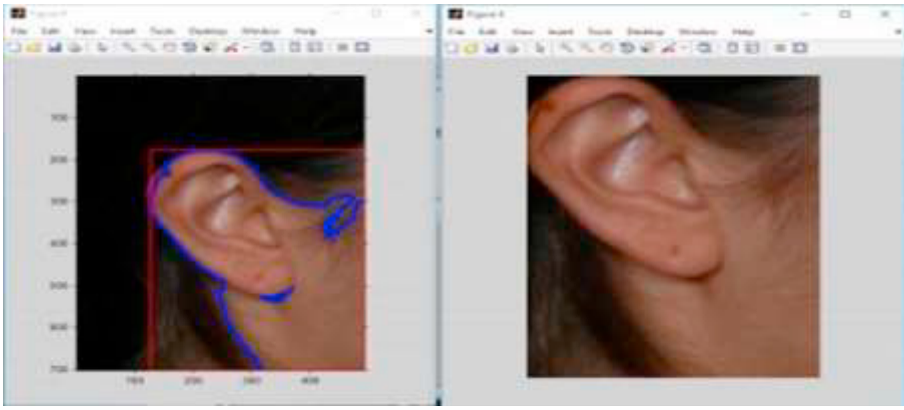

Fig 5. Segmentation of the ear

Now when a different image is given as the input, the above said process is repeated till the extraction of the SIFT features. In Fig 6., the output includes the LoG and gradient magnitude calculations. The right side of the image shows the position of the SIFT vectors that are detected for human ear. By using the Euclidean distance measure, the difference between the SIFT features of the two images is calculated and then the specific person is identified. This output is displayed as a pop-up window

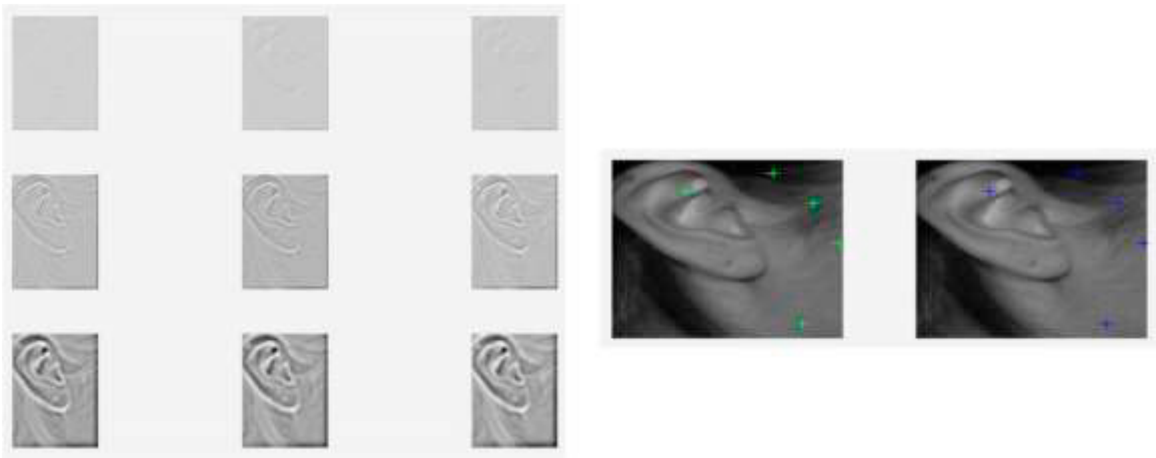

Fig 6. Extraction of SIFT features

5. Conclusion and future work

In this paper, a method for detection and recognition of the human ear for the use of bio metrics has been proposed and implemented. Images of human ear from various angles were collected and stored. Around 545 images were collected, of which 80% are used for training and the remaining 20% are used for testing purposes.

Initially the ear is detected and segmented from the input image using the Contour detection algorithm. The Scale Invariant Feature Transform (SIFT) has been applied on the segmented ear image in-order to extract the SIFT features. From the extracted features, the model file is generated from the training images. The test images carry out the same process and for recognition of the Ear, the Euclidean distance measure has been employed which gives the percentage of dissimilarity between the images. Thus using the value of this distance measure the specific person's ear has been identified. The further research will be on minimizing the feature values for better recognition .

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